Sheet 1: the simple model

1. $y_i = \beta + u_i$, $u_i \sim iid(0, \sigma^2)$, $\mathbb{E}[u_i^4] < \infty$

a) The OLS problem is

$$\min_{b \in \mathbb{R}} \sum_{i=1}^{n} (y_i - b)^2$$

1a. yes, correct

for which our FOC is $\frac{\partial L}{\partial b} = -2 \sum_{i=1}^{n} (y_i - b) \overset{!}{=} 0$, and solved by $\hat{\beta}$

Rearranging, $n\hat{\beta} = \sum_{i=1}^{n} y_i \Rightarrow \hat{\beta} = \frac{1}{n} \sum_{i=1}^{n} y_i = \bar{y}$ i.e. the sample mean.

b) $\sum_{i=1}^{n} \hat{u}_i \equiv \sum_{i=1}^{n} (y_i - \hat{\beta})$

$= \sum y_i - n\hat{\beta}$

1b. yes, correct

$= \sum y_i - n\bar{y}$ by (a)

$= 0$

c) We can rewrite $\hat{\beta} = \bar{y}$ as follows:

$= \frac{1}{n} \sum_{i=1}^{n} y_i = \frac{1}{n} \sum (\beta + u_i) = \frac{1}{n} \cdot n\beta + \frac{1}{n} \sum u_i$

$= \beta + \bar{u}$

So $\mathbb{E}[\hat{\beta}] = \mathbb{E}[\beta + \bar{u}]$

$= \beta + \mathbb{E}[\bar{u}]$ by linearity and since β is a param.

$= \beta$ as u_i is mean zero, so $E[\bar{u}] = \frac{1}{n} \sum E[u_i] = \frac{1}{n} \sum_{i=1}^{n} (0)$.

For variance, note that $var(\hat{\beta}) = var(\beta + \bar{u})$

1c. yup, all correct

$= var(\bar{u}) = \frac{1}{n^2} var(\sum u_i)$ by var rules

$= \frac{1}{n^2} \sum var(u_i)$ by zero autocorrelation of errors

$= \frac{1}{n} \sigma^2$ as iid

(expected, since $\hat{\beta}$ is just the sample mean)

[could use Chebyshev ineq. directly]

d) Yes, $\hat{\beta}$ is consistent. One ~~that~~ way to see this is that it's unbiased and its variance tends to zero as $n \to \infty$.

1d. yup, all correct!

More formally, the Chebyshev LLN states that where x_i is drawn iid with finite mean μ and variance σ^2, $\bar{x}_n \xrightarrow{p} \mu$. In this case, we have u_i drawn iid with mean 0 and variance σ^2, so $\bar{u} \xrightarrow{p} 0$.

good

$$\therefore \hat{\beta} = \beta + \frac{1}{n}\sum u_i \xrightarrow{p} \beta + 0 = \beta \text{ by Slutsky theorem.}$$

what would we (*) call this object?

e) Consider the standardised estimator (*) $\sqrt{n}(\hat{\beta} - \beta)$.

$$= \sqrt{n}(\beta + \bar{u} - \beta) = \sqrt{n}\,\bar{u}$$

1e. ok, correct!

$$= \sqrt{n}(\bar{u} - 0),$$ which (since u_i is iid w/ finite $\underset{0}{\underset{\shortparallel}{\mu}}, \sigma^2$) by Lindeberg-Levy CLT

$$\overset{a}{\sim} N(0, \sigma^2)$$

so $\hat{\beta}$ is asymptotically normal. ~~with an~~

f) You can calculate the t-statistic $t = \frac{\hat{\beta} - 0}{se(\hat{\beta})}$ (where $se(\hat{\beta})$ is a feasible estimate of $\sqrt{var(\hat{\beta})}$ to be defined) and then compare to critical values from the normal distribution.

~~First, we define $se(\hat{\beta})$~~ We know $\omega_{\hat{\beta}}^2 := var(\hat{\beta}) = \frac{1}{n}\sigma^2$, and we're interested in estimating this, since σ^2 is not known. Let $\hat{\sigma}^2$ be the error variance estimator $\frac{1}{n-1}\sum_{i=1}^{n} \hat{u}_i^2$. Then our $se(\hat{\beta}) = \sqrt{\hat{\sigma}^2/n} = n^{-1/2}\hat{\sigma}^2$, by (c).

$\hat{\sigma}^2$ is consistent for σ^2: $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1} (y_i - \hat{\beta})^2$

$$= \frac{1}{n-1}\sum(\beta + u_i - \beta - \bar{u})^2 = \frac{1}{n-1}\sum(u_i - \bar{u})^2$$

$$= \frac{1}{n-1}\sum(u_i^2 - 2u_i\bar{u} + \bar{u}^2) = \frac{1}{n-1}\left[\sum u_i^2 - n\bar{u}^2\right] = \frac{n}{n-1}\left(\frac{1}{n}\sum u_i^2\right) - \frac{n}{n-1}\bar{u}^2$$

~~as $n \to \infty$ We have $\lim_{n\to\infty}$~~ $\frac{n}{n-1}\bar{u}^2 = 1 \cdot 0^2 = 0$ by continuous mapping and u ~~mean zero~~

For the first term, as $n \to \infty$ we have

$\frac{n}{n-1} \cdot \frac{1}{n} \sum u_i^2 \to \frac{n}{n-1} \cdot \frac{1}{n} \sum$

f) To test $H_0: \beta = 0$ vs $H_1: \beta \neq 0$, you would calculate the test statistic $t = (\hat{\beta} - 0) / se(\hat{\beta})$ – where $se(\hat{\beta}) = \sqrt{\hat{\sigma}^2/n}$ is a feasible estimate of $\sqrt{var(\hat{\beta})}$ – and then compare its realised value to critical values from a normal distribution, rejecting H_0 iff $|t^{obs}| > c_\alpha$.

1f. well done, excellent answer!!

From (c) we have $\omega_\beta^2 := var(\hat{\beta}) = \frac{1}{n}\sigma^2$, but σ^2 is not known. Let our estimator of the error variance $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n} \hat{u}_i^2$, where $\hat{u}_i = y_i - \hat{\beta}$ as in (b). Note that $y_i - \hat{\beta} = (\beta + u_i) - (\beta + \bar{u})$, by (c).

Then $\hat{\sigma}^2 = \frac{1}{n-1}\sum(u_i - \bar{u})^2 = \frac{1}{n-1}\left[\sum u_i^2 - 2\sum u_i\bar{u} + n\bar{u}^2\right]$

$= \frac{1}{n-1}\left[\sum u_i^2 - n\bar{u}^2\right] = \frac{n}{n-1}\left[\underset{(1)}{\frac{1}{n}\sum u_i^2} - \underset{(2)}{\bar{u}^2}\right]$

~~Considering (1)~~ Now let $n \to \infty$. We have $\lim_{n\to\infty} \frac{n}{n-1} = 1$, and

(1) As u has finite fourth moment, we can apply the LLN to u^2, since $var(u^2)$ is defined. u is iid, thus so too is u^2, and hence $\frac{1}{n}\sum u_i^2 \xrightarrow{p} \mathbb{E}[u_i^2]$. And as $var(u) = \mathbb{E}[u^2] - (\mathbb{E}[u])^2$, and $\mathbb{E}[u] = 0$, $\mathbb{E}[u^2] = \sigma^2$.

(2) $\bar{u} = \frac{1}{n}\sum u_i \xrightarrow{p} 0 \; (= \mathbb{E}[u])$ by the LLN, and by continuous mapping $\bar{u}^2 \xrightarrow{p} 0$

So $\lim_{n\to\infty} \hat{\sigma}^2 = 1 \cdot (\sigma^2 + 0) = \sigma^2$, i.e. it is consistent.

$\therefore$ $se(\hat{\beta})$ is consistent for σ.

By the CLT, then, under H_0 $t = \frac{\sqrt{n}(\hat{\beta} - 0)}{\sqrt{\hat{\sigma}^2}} \xrightarrow{d} N(0,1)$ (as $\mathbb{E}[\hat{\beta}] = \beta \overset{H_0}{=} 0$ and $\hat{\sigma}^2 \xrightarrow{p} \sigma^2$, using Slutsky). This justifies the procedure above.

2. $y_i = \beta x_i + u_i$.

a) The OLS estimator $\hat{\beta}$ solves $\min_{b \in \mathbb{R}} \sum_{i=1}^{n} (y_i - bx_i)^2$, which has FOC $\frac{\partial L}{\partial b} = -2 \sum_{i=1}^{n} (y_i - bx_i) x_i \overset{!}{=} 0$.

2a. yes, correct!

Rearranging, $\sum_{i=1}^{n} (y_i x_i) = \hat{\beta} \sum_{i=1}^{n} x_i^2 \Rightarrow \hat{\beta} = \frac{\sum_{i=1}^{n} (y_i x_i)}{\sum_{i=1}^{n} x_i^2}$

b) $\sum_{i=1}^{n} \hat{u}_i x_i = \sum (y_i - \hat{\beta} x_i) x_i = \sum (y_i x_i - \hat{\beta} x_i^2)$

$= \sum y_i x_i - \hat{\beta} \sum x_i^2 = \sum y_i x_i - \frac{\sum y_i x_i}{\sum x_i^2} \cdot \sum x_i^2$

2b. yup, good answer!

$\underline{= 0}$

$\sum_{i=1}^{n} \hat{u}_i = \sum (y_i - \hat{\beta} x_i) = \sum y_i - \hat{\beta} \sum x_i \neq 0$ in general. Since we fit a slope-only regression (i.e. force the intercept to be zero), the sum of residuals need not equal zero in our sample even if $\mathbb{E}[x] = 0$.

~~Since we have $\mathbb{E}[x] = 0$, we can rewrite this as~~

c) $\mathbb{E}\left[\sum (y_i x_i) / \sum x_i^2\right] = \frac{\sum (\beta x_i + u_i) x_i}{\sum x_i^2} = \beta \frac{\sum x_i^2}{\sum x_i^2} + \frac{\sum u_i x_i}{\sum x_i^2}$

$= \beta + \frac{\sum u_i x_i}{\sum x_i^2}$ ~~$\Rightarrow \mathbb{E}[\hat{\beta}] = \beta + \mathbb{E}\left[\frac{\sum u_i x_i}{\sum x_i}\right]$ since $\mathbb{E}$~~

~~But we have $u_i \perp\!\!\!\perp x_i \Rightarrow \operatorname{cov}(u, x) = 0$~~, so $\mathbb{E}[\hat{\beta}] = \beta + \mathbb{E}\left[\frac{0}{\sum x_i^2}\right] = \beta$

~~hence it is unbiased.~~

So $\mathbb{E}[\hat{\beta} \mid X] = \beta + \mathbb{E}\left[\frac{\sum u_i x_i}{\sum x_i^2} \middle| X\right] = \beta + \frac{1}{\sum x_i^2} \sum x_i \mathbb{E}[u_i \mid X]$ by conditi...

$= \beta + \frac{1}{\sum x_i^2} \sum x_i \cdot 0$ (as $u \perp\!\!\!\perp x$ and mean zero)

$= \beta$, so unbiased.

Or alternatively since we condition on X, we can just use the fact that $u \perp\!\!\!\perp x \Rightarrow cov(u, x) = 0$

$var(\hat{\beta}|X) = var\left(\beta + \frac{\sum x_i u_i}{\sum x_i^2} \,\Big|\, X\right)$

and since both are mean zero, the numerator term $\mathbb{E}[u_i x_i] = 0$, and denominator is a constant.

$$var(\hat{\beta}|X) = \frac{var(\sum u_i x_i)}{(\sum x_i^2)^2}$$ ← constant as conditioning on X

$$= \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j \, cov(u_i u_j | X)}{(\sum x_i^2)^2}$$ by sum of variances and conditioning

For $i \neq j$, $cov(u_i u_j) = 0$ by assumption. 2c. yup, all good

$$= \frac{\sum_{i=1}^{n} x_i^2 \, var(u_i | X)}{(\sum x_i^2)^2} = \frac{\sigma^2}{\sum x_i^2}$$

d). Yes, $\hat{\beta}$ is consistent. $\hat{\beta} = \beta + \frac{\sum u_i x_i}{\sum x_i^2} = \beta + \frac{n^{-1}\sum x_i u_i}{n^{-1}\sum x_i^2}$

First consider the numerator. $(u_i x_i)$ are iid, thus so too is $u_i x_i$. By the LIE $\mathbb{E}[u_i x_i] = E[E[u_i x_i | x_i]]$

$= E[x_i E[u_i | x_i]] = E[x_i \cdot 0]$ by assumption of $u_i \perp\!\!\!\perp x_i$ and mean zero

$= 0$

and $\therefore var(u_i x_i) = E[E[u_i^2 x_i^2 | x]] = E[x^2 \cdot E[u^2 | x]]$

$= E[x^2]\sigma^2 < \infty$ 2d. excellent answer!

So by the Chebyshev LLN, $n^{-1}\sum x_i u_i \xrightarrow{p} \mathbb{E}[x_i u_i] = 0$

And the denominator $n^{-1}\sum x_i^2 \xrightarrow{p} \mathbb{E}[x^2] < \infty$ by LLN and since x has finite fourth moment.

So by Slutsky, $\hat{\beta} \xrightarrow{p} \beta + \frac{0}{\mathbb{E}[x^2]} = \beta$, assuming x has nonzero variance $\sigma_x^2 (=) \mathbb{E}[x^2]$

(given $\mathbb{E}[x]^2 = 0$)

e) Consider the standardised estimator $\sqrt{n}\,(\hat{\beta}-\beta) = \dfrac{n^{-1/2}\sum_i x_i u_i}{n^{-1}\sum_i x_i^2}$

We have from (d) that the denominator $n^{-1}\sum x_i^2 \xrightarrow{p} \sigma_x^2$.

For the numerator, we know $x_i u_i$ is iid, and $\mathbb{E}[x_i u_i] = 0$ from (d).
Further, $var(x_i u_i) = \mathbb{E}[x_i^2]\sigma^2 < \infty$ as they have finite fourth moment (and by a further result?)

since x_i and u_i are iid, their product x_i*u_i is also iid. You have worked out the mean and variance which are both finite. Hence, the CLT can be applied to the numerator

So we can apply the CLT: $n^{-1/2}\sum x_i u_i \xrightarrow{d} N(0, \sigma_x^2\sigma^2)$
as $n^{-1/2}\sum x_i u_i = \sqrt{n}\cdot\left(\frac{1}{n}\sum x_i u_i - \mathbb{E}[x_i u_i]\right)$

$\therefore \sqrt{n}\,(\hat{\beta}-\beta) \xrightarrow{d} \dfrac{N(0, \sigma_x^2\sigma^2)}{\sigma_x^2}$ by Slutsky

2e. yes, correct!

$= N(0, \sigma^2/\sigma_x^2)$

We just need to demonstrate again that $\hat{\sigma}^2 \xrightarrow{p} \sigma^2$ and thus we can feasibly estimate $se(\hat{\beta})$.

f) Similar to 1f), we calculate $t = \hat{\beta}/se(\hat{\beta})$ and compare to $N(0,1)$ CVs
Let the error variance estimator $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\hat{u}_i^2$, where $\hat{u}_i = (\beta x_i + u_i) - \left(\beta + \frac{\sum u_i x_i}{\sum x_i^2}\right)x_i$
$= \beta x_i + u_i - \hat{\beta} x_i$

~~Then $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(u_i - \frac{\sum u_i x_i}{\sum x_i^2} x_i\right)^2$~~

Then $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(u_i - (\hat{\beta}-\beta)x_i\right)^2$

$= \frac{n}{n-1}\Big[\underbrace{\tfrac{1}{n}\sum u_i^2}_{(1)} - \underbrace{2(\hat{\beta}-\beta)\tfrac{1}{n}\sum u_i x_i}_{(2)} + \underbrace{(\hat{\beta}-\beta)^2\tfrac{1}{n}\sum x_i^2}_{(3)}$

2f. partially correct, note that the standard error is (sigma_hat/sigma_x hat)/n . This can be seen from the asymptotic distribution that you correctly derived in the previous question. You correctly show that sigma^2_hat is a consistent estimator for sigma^2 (well done!)... but we also need to show that sigma^2_x hat is a consistent estimator for sigma^2_x

(1) $\frac{1}{n}\sum u_i^2 \xrightarrow{p} \sigma^2$ by (f).
(2) $(\hat{\beta}-\beta) \xrightarrow{p} 0$ by (d), and $\frac{1}{n}\sum u_i x_i \xrightarrow{p} 0$
(3) $(\hat{\beta}-\beta)^2 \xrightarrow{p} 0$ by continuous mapping, and $\frac{1}{n}\sum x_i^2 \xrightarrow{p}$ ...

So by Slutsky $\hat{\sigma}^2 \xrightarrow{p} \sigma^2 + 0\cdot 0 + 0\cdot\sigma_x^2$
$\sqrt{\hat{\sigma}^2/n}$ is consistent for σ, and hence by the CLT we have $t \sim N(0,1)$.